

packages/graph-core/src/search.ts

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Overview

`packages/graph-core/src/search.ts` adds a lightweight full-text search engine built on **Fuse.js**. It exports three symbols:

- `SearchResult` (R4-7) – `{ nodeId: string; score: number; /`
- `0 = perfect, 1 = worst */}``
- `SearchOptions` (R9-12) – `{ types?: GraphNode["type"][]; limit?: number; }`
- `SearchEngine` (R27-65) – a class that indexes an array of `GraphNode` objects.

Key changes

- **Imports:** `Fuse` and `IFuseOptions` from `fuse.js` (R1) and `GraphNode` from `./types.js` (R2).
- **Fuse configuration** (R14-25): keys with weights, `threshold: 0.4`, `includeScore: true`, `ignoreLocation: true`, `useExtendedSearch: true`.
- **SearchEngine:**
 - `constructor(nodes)` stores the nodes and builds a Fuse index.
 - `search(query, options?)` trims the query, turns space-separated tokens into an OR-style extended query (`"a b" → "a | b"`), runs Fuse, filters by `options.types` if supplied, limits to `options.limit` (default 50), and maps results to `SearchResult`.
 - `updateNodes(nodes)` rebuilds the internal index.

Impact

- **Correctness:** deterministic scoring (0–1) and filtering by type.
- **Maintainability:** all search logic resides in a single module; changes to `FUSE_OPTIONS` propagate automatically.
- **Compatibility:** introduces a runtime dependency on `fuse.js`; no existing public APIs are altered.

Risks & follow-ups

- **Dependency missing:** ensure `fuse.js` is listed in `package.json` and bundled.
- **Type mismatches:** `GraphNode` must expose `id`, `type`, `name`, `tags`, `summary`, and `languageNotes` as used in the Fuse keys.
- **Threshold tuning:** the chosen `0.4` may affect recall; benchmark if needed.
- **Extended search behavior:** the OR logic could match unintended tokens; validate with edge-case queries.